

Day 1 — How Markets Actually Work

Objective: Build the correct mental model of markets.

Time Required: ~40 minutes

Why are we doing this?

Before we write a single line of code, I want you to understand **what problem our algorithm is trying to solve**.

We're **not** trying to predict the future.

We're trying to make **better probabilistic decisions** than the average participant.

That distinction will shape everything we build.

The Biggest Myth About the Stock Market

Most people believe:

"A good company's stock always goes up."

This is false.

A better statement is:

A stock price reflects what the market collectively believes the company is worth *right now*.

Notice the difference.

The market prices **expectations**, not facts.

Think of the Market as an Auction

Imagine there is only one rare cricket bat signed by your favorite player.

Initially:

- One buyer offers ₹10,000.
- Another offers ₹11,000.
- Someone else offers ₹12,500.

Nobody changed the bat.

Only people's willingness to pay changed.

The stock market is the world's largest continuous auction.

Every second:

- Buyers compete.
 - Sellers compete.
 - The latest agreed price becomes the market price.
-

What Actually Moves Prices?

Think of price as being influenced by five forces:

Price

=

Information

+ Expectations

+ Supply & Demand

+ Liquidity

+ Human Behaviour

Let's understand each.

1. Information

Markets react to new information.

Examples:

- Earnings results
- Budget announcements
- RBI interest rate decisions
- Inflation data
- Wars
- Elections
- CEO resignations

Information changes what investors expect.

2. Expectations (Most Important)

This is where beginners usually go wrong.

Example:

A company reports:

Profit increased by **30%**.

Should the stock rise?

Not necessarily.

Suppose analysts expected **40%**.

The market may actually sell the stock because reality was **worse than expected**.

Markets compare:

Reality

vs

Expectation

—not just reality itself.

This single idea explains many "confusing" market reactions.

3. Supply and Demand

If:

100 people want to buy

10 want to sell

Price rises.

If:

100 want to sell

10 want to buy

Price falls.

Everything else eventually comes back to this imbalance.

4. Liquidity

Liquidity answers:

"How easy is it to buy or sell without moving the price?"

Example:

You own ₹5,000 of a highly traded stock.

Selling is easy.

Now imagine trying to sell ₹500 crore of the same stock.

That large order itself could push the price down.

This is one reason institutions behave differently from individuals.

5. Human Behaviour

Markets are emotional.

People experience:

- Fear
- Greed
- FOMO
- Panic
- Overconfidence

Even professionals aren't immune.

Our future platform exists to reduce emotional decision-making.

The Three Time Horizons

Different participants play different games.

Intraday Traders

Hold positions for minutes or hours.

Need speed.

Swing Traders

Hold for days or weeks.

Focus on trends.

This aligns well with your available time.

Investors

Hold for years.

Focus on business quality and long-term growth.

This complements your existing SIPs and equity portfolio.

Where Does Quantitative Trading Fit?

Traditional trader:

Looks at chart

↓

Feels bullish

↓

Buys

Quantitative trader:

Collects data

↓

Measures evidence

↓

Calculates probability



Checks risk



Decides

Emotion is replaced with process.

Our Goal

By the end of this journey, I don't want you asking:

"Should I buy this stock?"

Instead, I want you asking:

"What does my data say about the probability, expected return, and risk of this opportunity?"

That's a very different mindset.

Practical Exercise (≈40 minutes)

Part 1 (20 minutes)

Pick **three Indian stocks** and **three US stocks/ETFs**.

For each one, answer these questions without reading analyst opinions first:

1. Why do I think people are buying this today?
2. What expectations might already be reflected in the price?
3. What event could significantly change its price over the next three months?

Write down your answers. There are no "correct" responses here—the goal is to train your thinking.

Part 2 (20 minutes)

Create the folder structure for your project:

quant-platform/

```
|
|├── data/
|├── notebooks/
|├── src/
|  ├── data/
|  ├── indicators/
|  ├── scanner/
|  ├── strategies/
|  ├── portfolio/
|  └── backtesting/
|
|├── docs/
|├── tests/
|├── config/
└── README.md
```

In your README.md, write a one-page vision statement answering:

- What problem will this platform solve?
- Who is it for?
- What should it be able to do after one year?

This document will evolve alongside the platform.

Review Questions

Try answering these without looking back.

Question 1

A company announces record profits.

Its stock immediately falls 7%.

Which explanation is most likely?

- A. The company suddenly became weaker.
 - B. Investors expected even stronger results, so expectations were disappointed.
 - C. Stock prices always move randomly.
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Question 2

What is liquidity?

- A. The amount of cash a company has.
 - B. How easily an asset can be bought or sold without significantly affecting its price.
 - C. The number of shareholders.
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Question 3

Which statement best reflects a quantitative mindset?

- A. "This chart looks bullish."
 - B. "This stock feels cheap."
 - C. "Historically, setups with these characteristics produced positive expected returns after accounting for risk and trading costs."
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Homework

I'd like you to start a notebook called "**Quant Journal.**"

Every day, answer these four questions:

1. What did I learn today?
2. What surprised me?
3. What questions do I still have?
4. How could today's concept improve my future trading platform?

This journal will become one of your most valuable learning resources over the next year.